

AN INITIAL EXAMINATION OF SOUTHWEST SPANISH VOWELS

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ABSTRACT. This article investigates the Spanish vowel system of Southwest Spanish speakers through an acoustic examination of F1 and F2. The corpus is based on a semi-spontaneous narrative by four female speakers. Repeated measurements of all five Spanish vowels in a stressed syllable are plotted, as well as a comparison of 30 productions of /a/ in a stressed and unstressed syllable. The findings indicate several shifts in the generally accepted Spanish vowel triangle including a lowering and fronting of /u/, a lowering of /o/, and a fronting of /a/ to the vowel space typically described for English /æ/. There was no reduction of unstressed /a/ tokens to a schwa.*

INTRODUCTION. While English dialectal variation of segments has typically focused on differences in vowel production, dialectal and sociolinguistic studies of Spanish segmental variation have tended to concentrate on consonants (Zamora and Guitart 1982, Lipski 1994). This focus on vowels reflects Navarro Tomás' (1977) claim that Spanish vowels are more steady and firm than consonants and dialectal variation affects Spanish consonants more so than vowels. It is precisely these claims of invariability that are examined in the research presented here.

There are typically few studies of Spanish vowel variation, either of a historical nature as in the case of dialectal variation in the Spanish Peninsula, or of a synchronic perspective as motivated by language contact, e.g. in the case of indigenous languages. This study examines acoustically the Spanish vocalic system of fluent Southwest Spanish speakers (hereafter SWS) and compares the results with vowel system descriptions from seminal works on Spanish vowels. Specifically, the research presented in this study examines data from narrative speech and the formant values of the five contrastive vowels of SWS, as well as the role of lexical stress on the realization of /a/ in tonic and atonic positions.

*I am grateful to comments and suggestions by Scott Rushforth of New Mexico State University and three anonymous reviewers. Any errors remain my own.

The organization of the article is as follows: Section 2 reviews the literature on acoustic characterizations of the Spanish vowel system. Section 3 details the methodology for the study, including a discussion of laboratory speech versus more natural elicitation techniques. In Section 4 I present the findings on the acoustic examination of the SWS vowel system. Section 5 contains a discussion of the findings with respect to vowel variation in Spanish dialects, the role of stress in the realization of SWS vowels, and the role of contact in the variation observed. The conclusions of the study are reviewed in Section 6.

1. Review of literature. The Spanish vowel system has five contrastive vowels. Since Navarro Tomás (1977), it has generally been understood and accepted that the Spanish vowel system is typically stable. Navarro Tomás (1977) suggests there may be slight differences between stressed and unstressed vowels, but that the differences are likely below a perceptual level, and in no way similar to the type of reduction experimented by Catalan. This claim has been echoed by others with respect to reduction to schwa of unstressed vowels in the English vocalic system (Stockwell & Bowen 1965, Quilis & Esgueva 1983, Barrutia & Schwegler 1993).

1.1. SPANISH VOWEL SPACE. In most characterizations, the shape of the Spanish vowel system/space is roughly triangular, exemplified in Figure 1. Generative accounts of Spanish /a/, utilizing binary features of height and frontness, differ slightly from more descriptive characterizations due to differing ends. For example, the Spanish low vowel /a/ is typically described as (+back) in generative analyses (Cressey 1978, Núñez-Cedeño, Morales-Front, Prieto I Vives, & Hualde 1999), while more descriptive characterizations portray /a/ as low and central (Whitley 2002:28, Barrutia & Schwegler 1993).

1.2. ACOUSTIC ANALYSES. The few quantitative studies of Spanish vowels employing an acoustic formant analysis are based on laboratory speech, i.e., target vowels in carrier phrases (Phillips 1976, Méndez 1982, Quilis & Esgueva 1983, Godínez & Maddieson 1985, Martínez Celdrán 1995, Ladefoged 2001). Apart from Quilis and Esgueva (1983), which examines multiple Spanish dialects, most studies are on Peninsular Spanish (Bradlow 1995, Martínez Celdrán 1995). Exceptions to the Peninsular bias include data from six Tijuana Mexican speakers by Godínez (1981); Méndez (1982) examines a Puerto Rican dialect, but only the vowels /i, a, u/. Ladefoged (2001) also provides vowel formant data for one speaker, presumably of Los Angeles Spanish. The above studies are based on simple word lists or the target vowel produced within a carrier phrase similar to *Yo digo _____ para ti*, 'I say _____ for you'.

Table 1 lists the formant frequencies of the average of the cross dialectal sample of sixteen speakers reported by Quilis and Esgueva (1983), and data from five speakers of a Peninsular Spanish dialect by Bradlow (1995). The data are plotted in Figure 1 and are based on the mean formant values reported from each study.

	F1 VALUES			F2 VALUES		
	BRADLOW	QUILIS	DIFFERENCE	BRADLOW	QUILIS	DIFFERENCE
i	286	264	(22)	2147	2317	(170)
e	458	436	(22)	1814	1995	(181)
a	638	657	(19)	1353	1215	(138)
o	460	474	(14)	1019	888	(131)
u	322	293	(29)	992	669	(323)

TABLE 1. *Formant Values for F1 and F2 from Quilis and Esgueva (1983) and Bradlow (1995)*

The formant plot reveals some variation between the two reports. The Quilis and Esgueva (1983) characterization, identified by the tokens with an asterisk, presents a slightly more broadly distributed system than Bradlow (1995); however, aside from the /u/, the distributions of the two systems are relatively similar. In both of these studies the vowel tokens were elicited through a carrier phrase.

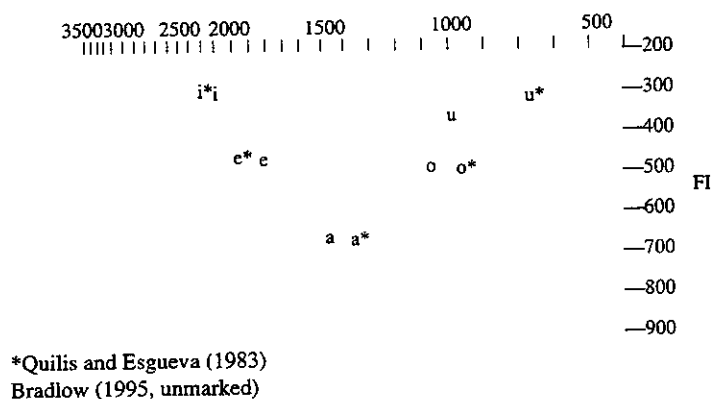


FIGURE 1. *Spanish Vowel Chart Plot*

1.3. DIALECTAL VARIATION OF SPANISH VOWEL QUALITY. The majority of the acoustic analyses of Spanish vowels within the last thirty years have focused on characterizing the general Spanish vowel system (Quilis and Esgueva 1983), or discussing an articulatory model (Godínez 1981). Barrutia and Schwegler (1993) claim that the differences found in the production of Spanish vowels are relatively insignificant across different dialectal regions. Lipski (1994) reports that there is some vowel reduction in atonic position among several dialects, including Mexican Spanish. However, tonic vowels are generally assumed to be produced in a uniform fashion across the many dialects. Studies that address Spanish vowel quality in the border region of the US and Mexico are based on perceptual analy-

ses of vowel production (Phillips 1967, Clegg 1969, Amastae 1978, Sánchez 1983). Despite some excellent work on Southwest and bilingual Spanish stop consonants (Zampini & Green 2001), currently our knowledge is limited concerning quantitative acoustic descriptions of Southwest Spanish vowels.

The literature review illustrates the lack of consideration or investigation of dialectal variation of the Spanish vowel system. Furthermore, the few acoustic analyses that have been conducted are based on laboratory speech. Finally, the review reveals significant gaps in our understanding of the impact of language contact and bilingualism on any vowel system of any dialect of Spanish, and in particular SWS.

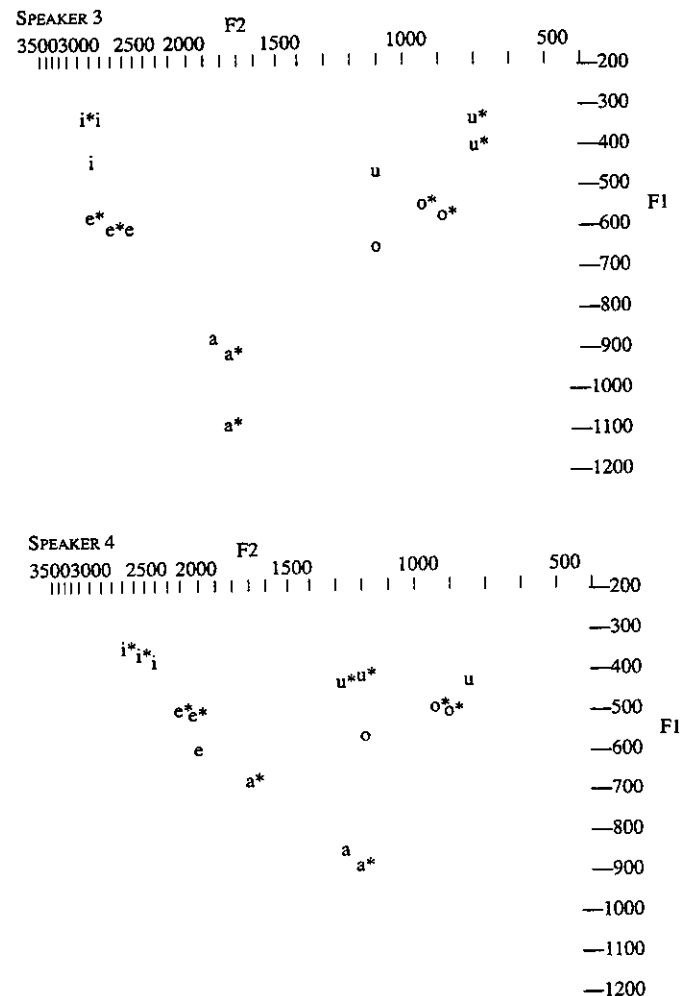
2. PROCEDURE. Informants for the current study were four bilingual female speakers from the US/Mexico border area (El Paso, Texas, and Anthony, New Mexico).¹ The speakers were all students at New Mexico State University, Las Cruces, New Mexico, and varied in age from their early twenties to mid thirties. The speakers were all students in an introductory linguistics class, but were recorded before the unit on phonetics and phonology. All speakers had originally learned Spanish in the home and typically began speaking English upon entering the elementary school system. They all spoke both English and Spanish in a self reported 40% to 60% combination from high school up to the time of the interview.

2.1. LABORATORY VERSUS NATURALISTIC SPEECH DATA. Harnegnies and Poch-Olivé (1992), in a study of two speakers, found significant variation in the production of vowels between spontaneous conversation and laboratory speech employing word lists extracted from previous spontaneous conversation. An initial pilot study of SWS vowels in isolated sentences tended to approximate the distributions described by Quilis and Esgueva (1983) and Bradlow (1995). However, these READ productions did not sound similar to the vowel quality observed in normal conversation with the students. This variation between read speech and vowels heard in normal conversations prompted a comparison of read speech versus naturalistic speech.²

There are striking differences in SWS vowel quality depending on the speech act as shown in Figure 2. The target sentences included two productions of each vowel in a stressed syllable. The narrative productions are the mean values from the narrative corpus for each vowel (5 productions of /i, e, o, u/, and 30 productions of the tonic /a/).

¹While not all sociolinguistic variables are controlled in the current study, the systematic patterning of these four speaker's vowels seen in Section 4 suggests a degree of homogeneity of the SWS system compared to other reports.

²For the current study, laboratory speech studies are typically those characterized by isolated words, or words produced in carrier phrases, 'I say ____ for you', described in previous studies on Spanish vowels. Spontaneous speech studies are those based on naturalistic communicative events (see e.g. Face 2003).



*Tokens with the asterisk indicate the read/laboratory productions

FIGURE 2. *Vowel Distributions of Read versus Narrative Speech*

In Figure 2 we can see that Speaker Three has a reduced and centralized vowel space in the natural data compared to the isolated laboratory style speech. Speaker Four also has a reduced vowel space in the narrative versus laboratory data, except for the vowel /u/. This variation in the vowel quality across the two types of speech act suggests that dialectal variation of Spanish vowels should be examined with more than laboratory style data.

Since the goal of the current study is to characterize a SWS dialect, it was decided that a more naturalistic speech act would achieve this goal more so than laboratory speech. Additionally, the trend in dialectal characterizations is toward spontaneous speech or at least both types for comparative purposes (Lewis 2001, Machuca-Ayuso 1997, O'Rourke 2005).

2.2. NARRATIVE DATA. The natural speech data was obtained from the telling of Mayer's (1969) picture story *Frog, where are you?* This is a book without words so there is no reading involved; rather, the informants produced a simple narrative telling of the story illustrated in the descriptive pictures (see Berman, Stromqvist, Slobin, & Verhoveen 1994 for additional information). The narrative from which the vowel tokens were extracted was part of a larger interview including their experience with Spanish, episodes of fear or happiness, a discussion about their family, and a small reading task. Each interview was conducted in Spanish and lasted approximately 20 minutes.³ The interviews were conducted in the researcher's office and digital recordings were made using a USBpre external sound card, CoolEdit 2000 sampled at 44k, and a head mounted Shure 512 boom microphone.

Only vowel tokens that were part of open syllables were examined to avoid potential variation due to a closed syllable. The vowel tokens were divided into two corpora designed to examine the general vowel system and evaluate the role of lexical stress on vowel quality. The vowel system characterization is based on the first five occurrences of each of the vowels /i, e, o, u/ in tonic position. The role of stress in the production of Spanish vowels was investigated with an examination of the vowel /a/.⁴ The data used in the analysis are the first thirty measurable tokens of each target vowel in the course of the narrative.⁵ The first set were /a/ tokens in an open stressed syllable in non-final position of the phrase,

³An anonymous reviewer questioned the naturalness of the recording setting. In response I can assert that all speakers were recorded in the same environment; therefore, the naturalness of the setting is equal for all informants.

⁴The vowel /a/ was chosen because it was the most frequently occurring vowel in the corpus and occurred in both stress positions.

⁵There were several types of vowel productions that were excluded from the data in the current study. Vowels productions not analyzed included glides with a sloped formant contour or vowels lacking a steady formant structure due to creaky or laryngealized voice (Ladefoged 2003:169-181). Additional studies are required to determine the degree and extent of these types of variations of productions.

illustrated by small capitals in (1). The second set of /a/ tokens used in the analysis are those tokens occurring in an unstressed open syllable in non-final position as illustrated by small capitals in (2).

- (1) El niño miraba la RANA.
The boy was looking at the frog.
- (2) El niño miraba la rANA saltar.
The boy was looking at the frog jump.

The acoustic signal was analyzed using the acoustic analysis software PRAAT (<http://www.fon.hum.uva.nl/praat/>).⁶ Formant frequencies of F1 and F2 were extracted from the central point of the vowel using the formant tracker. In general terms, F1 provides a measure of tongue position with respect to height based on the degree of constriction; the greater the constriction, the higher the vowel and the lower the F1 value. The F2 measurement is roughly correlated with the degree of frontness or backness. Back vowels have a lower F2 value compared to front vowels (Olive, Greenwood, & Coleman 1993:101-3).

Formant value comparisons of SWS with Quilis and Esgueva (1983), cited in Figures 3 and 4, are based on the reported values of the four Mexican speakers examined in their cross dialectal comparison, not on the average across all Spanish dialects reported in Table and Figure 1.⁷

3. Results.

3.1. Tonic vowel productions. The findings of the data indicate that the Spanish vowel system is not uniformly produced as repeatedly claimed in the literature (Navarro Tomás 1977, Quilis & Esgueva 1983, Barrutia & Schwegler 1993, Hualde 2005). The vowel quality of SWS reflected in F1 and F2 values is significantly different from the vowel values of Mexican speakers reported by Quilis and Esgueva (1983) as illustrated in Table 2 and Figure 3. The SWS /i/ vowel was considerably lower than the vowel formant values reported by Quilis and Esgueva (1983). The SWS values for /e/ were also lower than previous reports. The SWS /a/ was slightly lowered and fronted. The production of /o/ by the SWS speakers overlaps the Mexican vowel values /a/ reported by Quilis and Esgueva (1983), suggesting a phonetic/phonemic mismatch between the reports. Finally, the SWS /u/ was lowered and fronted compared to Quilis and Esgueva (1983).

⁶Formant frequencies were also confirmed with a spectrum analysis. The vowel formants were plotted with PlotFormant 4.0, developed at the UCLA Phonetics Lab.

⁷As can be noted, the Quilis and Esgueva (1983) formant values for all dialects cited in Table 1 and exemplified in Figure 1 differ slightly from the mean values of just the Mexican speakers in Figures 3 and 4 from the same study.

	F1 VALUES		F2 VALUES	
	QUILIS AND ESGUEVA	SWS	QUILIS AND ESGUEVA	SWS
i	264	431 (61)*	2317	2434 (204)
e	436	634 (73)	1995	2124 (161)
a	657	792 (125)	1215	1752 (124)
o	474	642 (112)	888	1197 (181)
u	293	450 (38)	669	1071 (198)

*Parentheses indicate the standard deviations

TABLE 2. *Comparative Formant Values for Quilis and Esgueva (1983) and the Current Combined Southwest Spanish Speakers' Data*

Figure 3 plots the vowel formant frequencies of the current SWS data and Quilis and Esgueva (1983) reports. The values from Quilis and Esgueva (1983) are listed in the graph with uppercase letters, while the SWS vowels are signaled with lowercase letters. The ellipse provides a representation of the distribution cloud and the dark circle next to an enlarged letter symbol indicates the mean. The distributional overlap of the SWS /o/ and the Quilis and Esgueva (1983) /a/ is clearly illustrated by the overlapping clouds and the proximity of the means.

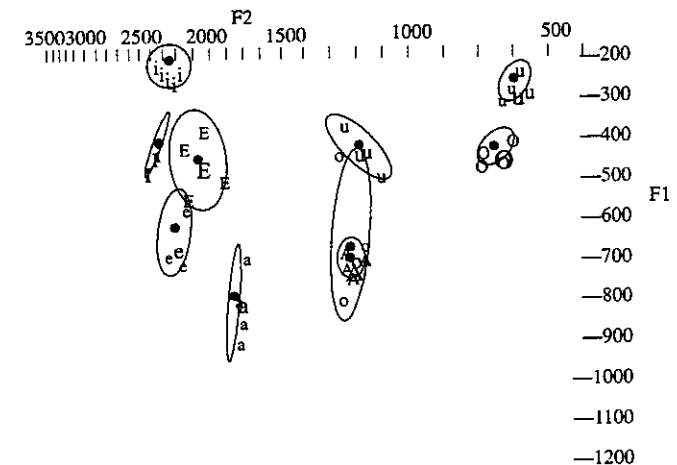


FIGURE 3. *Formant Plots for Mean Values of SWS Speakers and the Four Mexican Informants Formant Values Extracted from Quilis and Esgueva (1983)*

Except for the /o/ and /a/ overlap across the dialectal characterizations, each of the vowels tended to cluster in distinct distributions both within and across dialects.

3.2. TONIC VERSUS ATONIC /a/. Because SWS is a contact dialect and the informants were bilingual, there is a potential for English influence on the SWS system. This section compares SWS stressed and unstressed productions of the vowel /a/ to investigate the presence of vowel reduction, a hallmark of English. The mean values reported for each vowel are based on 30 productions of each. For three speakers, there was no significant difference between the tonic and atonic formant values (paired *t* Test). For speaker 3, there was a statistical difference between the tonic and atonic productions. However, productions of the vowels in both contexts fall within the distribution range of the other speakers, suggesting a noncategorical vowel quality variation.

	F1 VALUES			F2 VALUES		
	TONIC	ATONIC	DIFFERENCE	TONIC	ATONIC	DIFFERENCE
KW	786 (84)	731 (100)	55 (ns)	1771 (130)	1819 (153)	50 (ns)
RC	827 (84)	805 (63)	22 (ns)	1768 (77)	1783 (60)	15 (ns)
GM	890 (86)	823 (80)	67	1744 (97)	1785 (135)	41
RF	665 (120)	630 (69)	35 (ns)	1726 (176)	1726 (167)	0 (ns)

(ns) indicates no significant difference, paired *t* with significance level at .05

TABLE 3. *Formant Values and Differences between Tonic and Atonic /a/ Vowels in CV Syllables*

Figure 4 graphs the mean formant values of SWS tonic and atonic /a/ with the comparative values from the Mexican speakers in Quilis and Esgueva (1983). In Figure 4 we can see that there was no overlap in the formant plots of /a/ across the two dialects. The SWS vowels demonstrate a consistent degree of fronting and

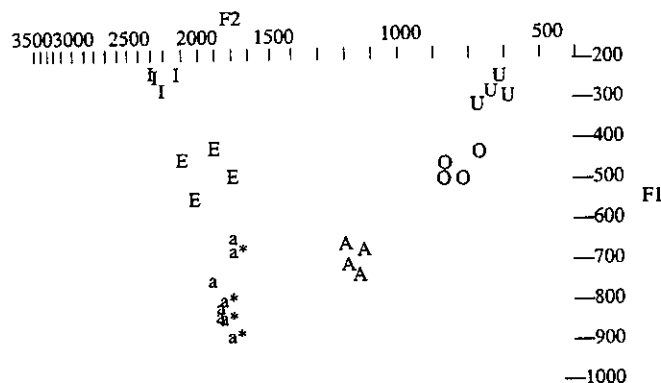


FIGURE 4. *Vowel Plot of Stressed and Unstressed /a/ Productions of SWS and the Mexican Vowel Values from Quilis and Esgueva (1983)*

lowering compared to the vowel values from the Mexican dialect, regardless of syllable stress. The stressed and unstressed vowel formants of SWS overlap in their distribution, suggesting that stress does not play a role in vowel quality production for this dialect.

4. DISCUSSION.

4.1. DIALECTAL VARIATION. The SWS vowel system differs significantly from the Spanish vowel system reported by Quilis and Esgueva (1983) and Bradlow (1995). Based on vowel plots of the Mexican speakers and SWS data, there were no overlapping distributions of phonemically identical vowel categories between the two dialects (see Figure 3). The only vowel distributions that were in an overlapping distribution belonged to separate phonemic categories, /o/ and /a/. The SWS /o/ and Mexican /a/ from Quilis and Esgueva (1983) had the same distribution of formant values. This similarity in phonetic production with distinct phonological categories demonstrates that Spanish vowel quality is not invariable in a naturalistic speech act as previously believed.

There were several additional differences of the SWS vowel system compared to the Mexican vowel system reported in Quilis and Esgueva (1983). One difference is that the SWS vowel system is more centralized than the Mexican system; however, as was seen in Figure 2, this variation may be due in part to the speech act type as described by Harnegies and Poch-Olivé (1992). SWS vowels are fronted compared to Mexican vowels, particularly the three non-front vowels. Additionally, the SWS front vowels /e/ and /i/ are also slightly more fronted than their Mexican vowel counterparts. A final characteristic of SWS is that the high vowels, /i/ and /u/, are significantly lower than previous reports, approximating the height of the standard Spanish midvowels /e/ and /o/. While this discussion has been limited to the Mexican vowel system reported in Quilis and Esgueva (1983), the results of the current study indicate there is more variation in vowel quality than previously believed.

4.2. THE ROLE OF STRESS ON VOWEL QUALITY. The presence of lexical stress does not appear to play a significant role in the production of the SWS /a/. In Figure 4 we can see an overlap of vowel space for the tonic and atonic vowel formant means. All /a/ vowel tokens were equally fronted. There may be a slight phonetic effect of lexical stress due to vowel height. However, much larger differences were observed in vowel height based on speech style (see Figure 2). The fact that vowel quality was affected more by speech type than by the phonological condition suggests that lexical stress does not exert an overt pressure on vowel quality in SWS based on the criteria examined.* There was no centralization or reduction of SWS unstressed /a/ to an English-like /ə/.

*The presence and type of pitch accent or boundary tone may play a role on vowel quality. This variable was not controlled for in the current study.

4.3. LANGUAGE CONTACT ISSUES. While the current study does not examine English vowels, evidence of contact influence in the Spanish to English direction provides evidence for vowel system permeability due to contact. For example, there are several studies of Chicano English with variations from the standard norm.⁹ Variations include the fronting of /ow/ and /uw/ and the lowering of /ae/ (Fought 1999), and the tensing, and raising of /I/ (Mendoza-Denton 1999). Godínez and Maddieson (1985) find statistically significant differences between Chicano English and Chicano Spanish with respect to vowel quality, but not quantity. These variations in English could be attributed in part to bilingualism and contact.

A hypothesized effect of English contact or transfer onto the Spanish system could be a tendency to reduce or centralize the atonic vowels to approximate the English schwa [ə]. However, as was illustrated in Figures 3 and 4, a mid-centralized vowel was not produced as a measurable variant of the /a/ tokens in an unstressed syllable. Also relevant is the lack of significant difference for the standard deviations of the two /a/ types. The lack of effect of lexical stress on the vowel quality of /a/ in SWS argues against a direct transfer of the English vowel system to these bilingual speakers.

While the SWS unstressed /a/ does not reduce to [ə], there are other, less direct possibilities for system contact. Figure 5 is a vowel plot of Standard American

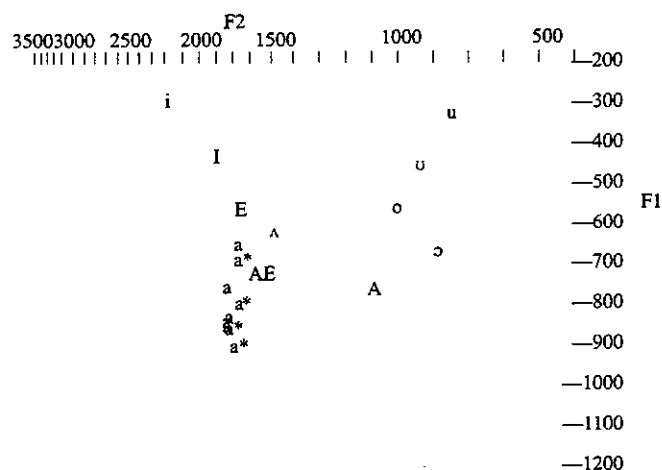


FIGURE 5. *Vowel Plot of Standard American English Vowels from Olive et al. (1993) Compared with the SWS Data /a/ Tokens, Identified with an Asterisk*

⁹This norm is never explicitly stated, though most general characterizations generally coincide (Olive et al. 1993, Ladefoged 2003).

English vowels from Olive et al. (1993), together with the mean values for the SWS /a/ tokens. From this comparison, we can see that the distribution of the SWS /a/ tokens overlaps the mean value of the Standard English /æ/ category.

The similarity in phonetic realization of SWS /a/ with English /æ/ may be due to language contact. However, other factors also need to be considered as possible motivators for the SWS /a/ fronting. For example, the SWS mid-vowel /o/ is lowered and fronted; similar to the Mexican /a/ vowel (see Figure 3). If the SWS /a/ were produced according to traditional accounts, there would be a potential vocalic neutralization of SWS /a/ and /o/. Therefore, the SWS /a/ fronting to a phonetic [æ] may be explained as a vowel push chain to maintain categorical distinctiveness of the two vowels. It is likely that both factors, English contact and vowel chain movement, are mutually influencing the current SWS vowel system. These questions require further investigation including additional sociolinguistic variables such as gender, age, and socioeconomic status, and dialectal variables such as monolinguals of both English and Spanish within the SWS dialectal region.

4.4. REPRESENTATIVENESS. The current speaker sample is small compared to most dialectal or sociolinguistic characterizations. However, the number of informants studied in the current characterization falls in the middle of the range of the normative studies for Spanish dialects (see section 2.2). It is certain that there are a number of factors, both phonological and sociolinguistic, that can influence vowel quality in a given dialect. However, the data from the current study patterns in a similar fashion suggesting some degree of representativeness.¹⁰ Furthermore, these SWS vowel quality patterns are significantly different from previous reports. While it is important to recognize that the previous reports used for comparison are based on a different type of speech, the vowels are significantly different compared to vowel characterizations in other reports.

5. CONCLUSION. The current study challenges long held assumptions concerning the relative stability of vowels across the dialects of Spanish based on naturalistic speech and using vowel formant data. This study finds that the border bilingual Spanish vowel system is different from the standard characterizations of the Spanish vowel system (Quilis & Esgueva 1983). The current findings indicate that the SWS vowel space is reduced in a naturalistic speech act compared to isolated read sentences, similar to the findings by Harmegnies and Poch-Olivé (1992). SWS vowels are fronted compared to vowels reported by Quilis and Esgueva (1983). The SWS /o/ vowel is lower and fronted to the vowel space described for Standard Spanish /a/, and the SWS /a/ is significantly fronted to the vowel space for /æ/

¹⁰The reduced number of informants is consistent with the literature on laboratory phonology. For example, Herrick (2003) based his work in Catalan on the speech of three females from different dialects in a comparison of vowel reduction.

described in Olive et al. (1993). Lexical stress does not play a significant role on the vowel quality of /a/ in SWS; unstressed /a/ did not reduce to schwa. This study concludes that there is vowel quality variation across certain Spanish dialects.

REFERENCES

- AMASTAE, JON. 1978. The acquisition of English vowels. *Papers in Linguistics* 11.423-57.
- BARRUTIA, RICHARD, and ARMIN SCHWEGLER. 1994. *Fonética y fonología españolas: Teoría y práctica*. New York, NY: Wiley.
- BERMAN, RUTH A.; DAN SLOBIN; SVEN STROMQVIST; and LUDO TH. VERHOEVEN. 1994. *Relating events in narrative: A crosslinguistic developmental study*. Hillsdale, NJ: Lawrence Erlbaum.
- BRADLOW, A. R. 1995. A comparative acoustic study of English and Spanish vowels. *Journal of the Acoustical Society of America* 97.1916-24.
- FACE, TIMOTHY. 2003. Intonation in Spanish declaratives: Differences between lab speech and spontaneous speech. *Catalan Journal of Linguistics* 2.115-31.
- FOUGHT, CARMEN. 1999. A majority sound change in a minority community: /u/ fronting in Chicano English. *Journal of Sociolinguistics* 3.5-23.
- CLEGG, JOSEPH HALVOR. 1969. *Fonética y fonología del español de Texas*. Austin, TX: University of Texas at Austin dissertation.
- CRESSEY, WILLIAM W. 1978. *Spanish phonology and morphology: A generative view*. Washington, DC: Georgetown University Press.
- GODÍNEZ, MANUEL. 1981. An acoustic study of Mexican and Brazilian Portuguese vowels. *Hispania* 64.594-600.
- GODÍNEZ, MANUEL, and IAN MADDIESON. 1985. Vowel differences between Chicano and General Californian English. *International Journal of the Sociology of Language* 53.43-58.
- HARMEGNIES, BERNARD, and DOLORS POCH-OLIVÉ. 1992. A study of style-induced vowel variability: Laboratory versus spontaneous speech in Spanish. *Speech Communication* 14.29-37.
- HERRICK, DYLAN. 2003. *An acoustic analysis of phonological vowel reduction in six varieties of Catalan*. Santa Cruz, CA: University of California at Santa Cruz dissertation.
- HUALDE, JOSÉ IGNACIO. 2005. *The sounds of Spanish*. Cambridge: Cambridge University Press.
- LADEFOGED, PETER (ed.) 2001. *A course in phonetics* (4th edition). Fort Worth, TX: Harcourt College Publishers.
- LADEFOGED, PETER. 2003. *Phonetic data analysis: An introduction to fieldwork and instrumental techniques*. Malden, MA: Blackwell Publishing.
- LEWIS, ANTHONY M. 2001. *Weakening of intervocalic /p,t,k/ in two dialects of Spanish: Toward the quantification of lenition processes*. Urbana-Champaign, IL: University of Illinois at Urbana-Champaign dissertation.
- LIPSKI, JOHN M. 1994. *Latin American Spanish*. New York, NY: Longman.
- MACHUCA AYUSA, MARÍA JESÚS. 1997. *Las obstruyentes no continuas del español: Relación entre las categorías fonéticas y fonológicas en habla espontánea*. Barcelona, Spain: Universitat Autònoma de Barcelona dissertation.
- MARTÍNEZ CELDRÁN, EUGENIO. 1995. En torno a las vocales del español: Análisis y reconocimiento. *Estudios de Fonética Experimental* 7.195-218.
- MAYER, MERCER. 1969. *Frog, where are you?* New York, NY: Dial Press.
- MÉNDEZ, ANTONIO. 1982. Production of English and Spanish vowels. *Language and Speech* 5.191-97.
- MENDOZA-DENTON, NORMA. 1999. Sociolinguistics and linguistic anthropology of US Latinos. *Annual Review of Anthropology* 28.375-85.
- NAVARRO TOMÁS, TOMÁS. 1977. *Manual de pronunciación española* (19th edition). Madrid: Consejo Superior de Investigaciones Científicas Instituto Miguel de Cervantes.
- NÚÑEZ-CEDENO, RAFAEL; ALFONSO MORALES-FRONT; PILAR PRIETO I VIVES; and JOSÉ IGNACIO HUALDE. 1999. *Fonología generativa contemporánea de la lengua española*. Washington, DC: Georgetown University Press.
- OLIVE, JOSEPH; ALICE GREENWOOD; and JOHN COLEMAN. 1993. *Acoustics of American English speech: A dynamic approach*. New York, NY: Springer-Verlag.
- O'ROURKE, ERIN. 2005. *Intonation and language contact: A case study of two varieties of Peruvian Spanish*. Urbana-Champaign, IL: University of Illinois at Urbana-Champaign dissertation.
- PHILLIPS JR., ROBERT NELSON. 1967. *Los Angeles Spanish: A descriptive analysis*. Madison, WI: University of Wisconsin dissertation.
- PHILLIPS JR., ROBERT NELSON. 1976. The segmental phonology of Los Angeles Spanish. *Studies in Southwest Spanish*, ed. by Donald Bowen and Jacob Ornstein, 74-92. Rowley: Newbury House Publishers.
- QULIS, ANTONIO, and MANUEL ESGUEVA. 1983. Fonemas vocálicos españoles. *Estudios de fonética*, ed. by Manuel Esgueva and Margarita Cantarero, 159-252. Madrid: Consejo Superior de Investigaciones Científicas Instituto Miguel de Cervantes.
- SÁNCHEZ, ROSAURA. 1983. *Chicano discourse: Socio-historic perspectives*. Rowley, MA: Newbury House Publishers.
- STOCKWELL, ROBERT P., and J. DONALD BOWEN. 1965. *The sounds of English and Spanish: Contrastive structure series*. Chicago, IL: University of Chicago Press.
- WHITLEY, M. STANLEY. 2002. *Spanish/English contrasts: A course in Spanish linguistics*. Washington, D.C.: Georgetown University Press.
- ZAMORA, JUAN, and JORGE GUITART. 1982. *Dialectología hispanoamericana: Teoría, descripción, historia*. Salamanca: Almar.
- ZAMPINI, MARY L., and KERRY P. GREEN. 2001. The voicing contrast in English and Spanish: The relationship between perception and production. *One mind, two languages: Bilingual language processing*, ed. by Janet L. Nicol, 23-48. Malden: Blackwell Publishers.